Chemistry 3A – Fall 2025 WITH EXPLANATIONS **Version A**

Sections 43957-43958

Midterm Examination #3

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which of these is important for gases to be considered ideal, and for the ideal gas law equation to work?

a) the gas molecules must be packed closely together
b) intermolecular attractions are very important for the molecules

c) gases should be able to easily fill a container

d) kinetic energy or the energy dependent on motion is not a consideration for understanding gases

Even if you are not sure about choice (c), the other three choices are false and go against Kinetic Molecular Theory and the assumptions of what is necessary for an ideal gas

2. Which of the following is true about pressure as related to gases?

a) Pressure is force exerted by gas molecules colliding with a surface over a given area (*P* = *F*/*A*)

b) Pressure is always (only) measured in units of atmospheres (atm)
c) Pressure is defined as the number of moles of gas present in a container
d) Pressure is a measure of the total energy of gas molecules in a container

There should be *one* TRUE choice and *three* FALSE choices. Choice (b) is FALSE because pressure has many units other than atmosphere (atm) as discussed: millimeters of mercury (mmHg) and torr have been mentioned, as well as others both metric and non-metric. Choice (c) shows a definition of pressure NOT given, and pressure is NOT a measure or indicator of energy of gas molecules. Choice (a) has been in fact given in lecture.

3. Convert a pressure of **595 torr** to atmospheres (**atm**)

a) 0.783 atm b) 1.28 atm c) 0.0595 atm d) 7.83 atm

Application of conversion factors for pressure is a fundamental calculation in basic chemistry

$$595 \text{ torr} \times \frac{1 \text{ atm}}{760 \text{ torr}} = 0.783 \text{ atm}$$

4. Which of these shows that the volume of a gas is directly proportional to the amount of a gas in moles ($V = k \times n$) ?

a) Boyle's Law b) Charles' Law c) Gay-Lussac's Law
d) Avogadro's Law e) Ideal Gas Law

Avogadro, famous for the mole concept, would likely be the one to show a law that explains how gas volume relates to the amount of gas (this is AMOUNT, which is typically measure in moles, not MASS, typically measured in grams. How do you relate amounts of substance to mass of substances, that is moles to grams? Yes, the molar mass)

5. A gas has an initial pressure P_1=10.0 atm and volume V_1 = 10.0 L. The pressure is reduced to P_2 = 1.00 atm. What is new volume V_2 in liters (L) if all else (temperature, moles) is constant?

a) 0.100 L **b) 50.0 L** **c) 10.0 L** **d) 1.00 L**

FIRST: this question was set up differently originally and the choices given were correct at the time. But when you calculate the problem, it is clear that NONE of the choices has the correct answer. This was corrected at the time of the exam for the students. Choice (b) was set at "100. L"

Pressure and volume are mentioned and a calculation is requested. The student who studied knows this is Boyle's Law, where *PV = constant (k)*. The math is $P_1V_1 = P_2V_2$, and usually P_1 and V_2 are the initial conditions, and you look for other P_2 or V_2 as the final conditions. The algebra is about selecting P_2 or V_2 depending on what is asked for. In this case, the volume V_2 is requested: $V_2 = \frac{P_1V_1}{P_2} = \frac{(10.0\text{ atm})(10.0\text{ L})}{(1.00\text{ atm})} = 100\text{ L}$

This should have been choice (b) in a correctly written (non-typo) exam.

6. An ideal gas of n = 1.00 mol at a temperature T = 273 K is contained in a volume V = 100.0 L. What is the pressure in atmosphere (atm)?

a) 0.0224 atm **b) 0.224 atm** c) 2.24 atm d) 22.4 atm

The question brings up the ideal gas so the ideal gas law equation should be used. It asks to solve the pressure P part of the equation, so that algebra should be used. The Yellow Sheet value for R will be used → 0.082058 L atm/(mol K). This form of R constant enables the correct cancellation of units used in the input values: mol, K, and L (!!) You have to learn to recognize that.

$$\boldsymbol{PV = nRT} \rightarrow \frac{\boldsymbol{nRT}}{\boldsymbol{V}} = \frac{(1.00\text{ mol})\left(0.082058\left[\frac{\text{L atm}}{\text{mol K}}\right]\right)(273\text{ K})}{(100.0\text{ L})} = 0.224\text{ atm}$$

7. An aqueous solution of sodium sulfate (Na_2SO_4) is prepared. Which of these is a ***true*** statement about this solution?

a) Sodium sulfate (Na_2SO_4) is the *solvent* and water (H_2O) is the *solute*

b) In nonsaturating aqueous solutions, sodium sulfate dissolves fully to form ions Na^+ and SO_4^{2-}

c) It is impossible to Na_2SO_4 dissolve in water because water will not dissolve ionic compounds

d) Sodium sulfate will form a precipitate and precipitates are not visible in solutions

So this is again a set of four true-false statement wrapped in a multiple choice question. Choice (a) is false as the compound sodium sulfate is the solute and water is typically the solvent. Choice (c) is also false: Na2SO4 is an ionic compound, and these generally dissolve in polar solvents like water. If you check the Green Sheet, a compound like sodium sulfate dissolves in abundant amounts in water: it will NOT form a precipitate and the additional false statement of "precipitates are not visible in solution" is the very opposite of what precipitates are, which make solutions cloudy in aqueous solution because they are insoluble and will settle out. Choice (b) is a true statement since nonsaturating solutions means that the solid compound completely dissolves in aqueous solutions.

8. Which of the statements below is a true statement regarding *electrolytes*?

a) The ionic solute dissolving will make a very good conductor of electric current
b) The ionic compounds that dissolve in solution do so (almost) entirely
c) Molecules like ethanol and glucose make good electrolytes

d) Both (a) and (b)

Choices (a) and (b) are true. Electrolytes require ions in aqueous solution, and ionic compounds in solid form must dissolve in solution to some extent. Choice (c) is false since ethanol (a liquid) and glucose (a solid), although they dissolve (or mix) will water in aqueous solution, do not form ions (are not ionic), and electrolytes are solutions of ions.

9. Which of the statements below refer to the meaning of "like dissolves like"?

a) It refers to combustion reactions with oxygen in the atmosphere
b) It means that the compound calcium fluoride (CaF_2) will dissolve in a nonpolar solvent like octane (C_8H_{18})

c) It means polar solutes will dissolve in polar solvents

d) It means that gases whose volume is reduced by half will see their pressures reduced by half

[**NOTE: the original question when printed did not have choices that cut across the page break.]**

"Like dissolves like" means nonpolar solutes prefer nonpolar solvents, and polar (or ionic) solutes prefer polar solvents. Choice (a) has nothing to do with this, as it discusses combustion. Choice (b) is the very opposite of like dissolves like. Choice (d) is another distractor which has nothing to do with the concept. Choice (c) exactly describes the concept.

10. Which of these solutes will have a greater effect on elevating the boiling point temperature of a solvent given they completely dissolve?

(a) ethanol (CH_3CH_2OH) b) glucose ($C_6H_{12}O_6$) c) NaCl **d) $(NH_4)_2SO_4$**

The question should have been understood quickly has a colligative properties question, where solutes have effects on properties of the solvent. In increasing the boiling point or decreasing the freezing/melting point temperature, it has been determined that the number of solute particles have the greater effect. This means that a compound that produces the most particles will have the greatest effect. Ethanol and glucose form only one particle in solution (aqueous solvent), NaCl will form two particles (two ions), and ammonium sulfate ($(NH_4)_2SO_4$) will form three. Choice (d) is correct.

11. A dilution of a C_1 = 0.50 *M* NaCl stock solution was used to make a V_2 = 500 mL of a C_2 = 0.10 M NaCl solution. What volume aliquot V_1 was used?

a) 20 mL b) 50 mL **c) 100 mL** d) 250 mL

One of the concepts in the lecture on solutions was about dilutions, and the relevant math is $C_1V_1 = C_2V_2$, where C_1 and C_2 are the initial (stock solution) and final (dilution) concentrations, respectively, and the V_1 and V_2 are the corresponding volumes. Keep in mind this is a conservation of mass equation: when you multiply concentration of a solute in aqueous solution with the volume of the solution, you get the mass of the solute. In this case we are seeking the volume used from the original stock concentration.

$$V_1 = \frac{C_2\boldsymbol{V}_2}{C_1} = \frac{(0.10\ M)(500\ \text{mL})}{(0.50\ M)} = 100\ \text{mL}$$

12. Which of the statements below are true about written chemical reactions?

a) Conservation of mass is not a consideration in the balancing of chemical reactions

b) The chemical compounds to the left of the arrow are called "reactants"

c) The chemical compounds to the left of the arrow are called "products"

d) The number and type of atoms on both sides of the reaction arrow (⟶) might NOT be equal

So another evaluation of four true-false choices. Choice (a) has been repeatedly stated to be false. The yield symbol ("reaction arrow") in chemical reaction/equations basically shows and separates what are the reactants, to the left of the arrow, and the products, to the right of the arrow. Choice (c) does not fit that. And choice (d) another way of stating mass is not conserved in chemical reactions, which is false. Choice (b) is the only correct choice.

13. In written chemical reaction equations, which of these symbols refers to a *gaseous* phase?

a) (*s*) b) (*l*) **c) (g)** d) (*aq*) e) none of these

[NOTE: this question had five choices. This was the first semester the instructor was giving this course after a return to teaching, so the educational policy of the instructor did not make firm decisions on many matters, including the format of summative assessments (testing).]

This question deals with the concept of "phases" (states of matter) observed in chemistry: *solid* ***(s)***, *liquid* ***(l)***, *gas(eous)* ***(g)***, *and aqueous* ***(aq)***

14. Precipitation reaction $Pb(NO_3)_2$ (*aq*) + NaCl (*aq*) ⟶ $PbCl_2$ (*s*) + $NaNO_3$ (*aq*) is shown. What is true about balancing this reaction?

a) Balancing requires changing the chemical formula of NaCl to $NaCl_2$

b) The reaction equation should only ever be balanced by adjusting the whole-number coefficients of the compounds

c) Removing NaCl from the reactants would make the equation balanced

d) Balancing requires changing $PbCl_2$ from solid (*s*) to aqueous (*aq*) to match other products

Choice (a) should be seen as wrong because there is no compound "$NaCl_2$" ever indicated in this chemistry course. Choice (c) is wrong because the balancing of reactions has been stated that balancing is not done by manipulating the chemical compound formulas themselves. Choice (d) is wrong because the phases of the compound have never been involved in balancing. Choice (b) is correct because the statement on balancing is that it is done by adjusting coefficients of the compounds in the reaction.

15. Which choice show pattern of a decomposition type reaction, e.g. 2 HgO (*s*) ⟶ 2 Hg (*l*) + O_2 (*g*)?

a) A + B ⟶ AB

b) AB ⟶ A + B

c) AB + CD ⟶ AC + BD

d) A + BC ⟶ AC + B

Scheme 1 reaction types are basically given in the choices. Choice (a) is the combination type, where two compounds combine to form one compound. Choice (c) looks like double replacement/displacement type, and Choice (d) is the single replacement/displacement type. None of those seems to be the type of the reaction in question, which is a decomposition type and fits the template or pattern of choice (b)

16. Which reaction below shows a **halogen replacement** (single replacement) reaction type?

a) Cl_2 (*g*) + 2 NaBr (*aq*) ⟶ 2 NaCl (*aq*) + Br_2 (*l*)

b) Mg (*s*) + $Cu(NO_3)_2$ (*aq*) ⟶ $Mg(NO_3)_2$ (*aq*) + Cu (*s*)

c) Zn (*s*) + 2 HCl (*aq*) ⟶ $ZnCl_2$ (*aq*) + H_2 (*g*)

d) H_2O (*l*) ⟶ H_2 (*g*) + 2 O_2 (*g*)

Choice (b) looks like the metal-type single-replacement, where a more "active" metal like Mg (magnesium) replaces Cu (copper). Choice (c) looks like a gas-evolving redox single replacement type. Choice (d)—which is not even balanced!!—looks like a decomposition, the electrolysis of water (even if choice (d) were true, it is still false because of the imbalance!). Choice (a) is one more reactive halogen (Cl_2) replacing a less reactive halogen (Br_2)

17. Which of these reactions of barium chloride ($BaCl_2$) and sodium sulfate (Na_2SO_4) shows the **net ionic equation**?

a) $BaCl_2\ (aq) + Na_2SO_4\ (aq) \longrightarrow BaSO_4\ (s) + NaCl\ (aq)$

b) $Ba^{2+}\ (aq) + 2\ Cl^-\ (aq) + 2\ Na^+\ (aq) + SO_4^{2-}\ (aq) \longrightarrow BaSO_4\ (s) + 2\ Cl^-\ (aq) + 2\ Na^+\ (aq)$

c) $Ba^{2+}\ (aq) + SO_4^{2-}\ (aq) \longrightarrow BaSO_4\ (s)$

d) all of them show the form called the **net ionic equation**

Remember those three chemical equations you need to know? The **complete chemical equation (**better and now called **molecular equation**), the **complete ionic equation**, and the **net ionic equation**? Those are choices (a), (b) and (c) respectively. Choice (c) is of the correct informative form. Choice (d) is a distractor.

18. 3.00 mol of potassium chloride (KCl) [molar mass = 74.55 g/mol] is dissolved in final 3.00 kg water. What is the **molality** of the KCl solution?

a) **1.00 m** b) 5.00 m c) 24.85 m d) 50.0 m

Molality—not molarity—is the moles of a solute divided by the kilograms of the solvent (not the solution). So you need to know the moles of a solute and the mass in kg of the solvent. The solute is KCl, and there are 3.00 mol. That solves the one quantity. Water is the solvent—both KCl and water will form the solution!—and there is 3.00 kg. So apply the calculation: 3.00 mol KCl / 3.00 kg water = 1.00 *m* KCl (note the lowercase italicized *m*) for the units. Why the molar mass of KCl was given? It was a distractor, to see if the student would know what is going on.

19. This is written on the whiteboard: "$Mg \longrightarrow Mg^{2+} + 2\ e^-$". What is true about what is written?

a) This shows a reduction half-reaction because Mg is gaining electrons

b) This shows is a gas as a product of the reaction

c) Shows an oxidation half-reaction because Mg went from a zero charge state to a +2 charge state

d) The number of protons should be shown on the left side of the half-reaction

Looking for the true choice, so we have four T-F choices to analyze. Choice (a) is false since the what is shown is an oxidation half reaction, not reduction, and electrons are actually lost. Choice (b) doesn't show any product that is a gas; this is a distractor. Choice (d) is false, since there is absolutely no requirement that these half reactions require H^+ to be shown in the reaction. Choice (c) describes what is being seen.

20. 100.0 g of ammonium chloride (NH_4Cl) [molar mass = 53.50 g/mol] is dissolved in final 2.00 L water. What is the concentration of the NH_4Cl solution?

a) 0.0374 M b) 0.2675 M **c) 0.935 M** d) 3.74 M

This asks for a molarity calculation using its definition. Molarity is moles solute / L solution (as a volume). The mass and not moles is given, so a conversion should be set up:

$$\frac{100.0\ \text{g NH}_4\text{Cl}}{2.00\ \text{L solution}} \times \frac{1\ \text{mol NH}_4\text{Cl}}{53.50\ \text{g NH}_4\text{Cl}} = 0.934579\ \text{mol/L} = 0.934579\ \text{M} \rightarrow 0.935\ \text{M}$$

21. What statements are true concerning the **oxidation number** of molecule carbon dioxide (CO_2)?

a) The oxidation number of **oxygen** (O atoms) is **–2**
b) The oxygen atoms in the molecule are more **electronegative** than for carbon
c) The oxidation number of **carbon** (the C atom) is **+4**

d) statements (a), (b) and (c) are all true

e) none of the statements above are true

Testing the knowledge of oxidation number. There are five choices to analyze. Choice (a) is true in that it is a rule and it applies to CO_2. Choice (b) is also true (I think we can see where this is going). Choice (c) also true. That seems to make choice (d) the correct choice and choice (e) is eliminated. Future exams will generally not make use of "all of the above" or "none of the above" style choices, as they can be deceptive.

22. Which statement is LEAST relevant or is INCORRECT regarding colligative properties of a solution?

a) These properties depend only on the **numbe**r of solute particles dissolved in the solvent.
b) **Molality** (*m*) is the required unit of concentration for calculating freezing and boiling point changes.
c) A solution's change in boiling point and freezing point are examples of colligative properties

d) To understand the effect of the solute, you must know the oxidation numbers of the atoms in the compound.

Tests knowledge of colligative properties. Choice (a) is correct about these properties. Choice (b) is actually used in these temperature calculations. Choice (c) is a correct statement. Since these first three choices actually apply, choice (d) would seem to be incorrect or false, and this is true since there is no need to know oxidation numbers and to understanding how colligative properties are utilized.

23. Lithium (Li) metal will react explosively with cold water in a vigorous redox reaction. Which of the reactions below is correctly balanced and shows what occurs?

a) 2 Li (*s*) + 2 H_2O (*l*) ⟶ 2 O_2 (*g*) + 3 LiOH (*aq*)
b) Li (*s*) + 3 H_2O (*l*) ⟶ H_2 (*g*) + LiOH (*aq*)
c) 2 Li (*s*) + 2 H_2O (*l*) ⟶ H_2 (*g*) + 2 LiOH (*aq*)
d) Li (*s*) + H_2O (*l*) ⟶ O_2 (*g*) + LiOH (*aq*)

Best way to solve this is to see which one reaction is balanced with the other three being unbalanced. Choice (a) is immediately eliminated since quick inspection shows 2 Li to left of arrow and 3 Li on right. Choice (b) is eliminated because there are 6 H atoms on left of arrow and 3 on the right. Choice (c) looks good but let's check (d) first: There are 3 O atoms on right, and only one O atom on left. Let's verify choice (c): 2 Li atoms on both sides, 4 H atoms on both sides, and two O atoms on both sides, so we're good.

24. What happens to the solubility of most gases in water as temperature increases?

a) It increases **b) It decreases** c) It remains constant
d) It fluctuates randomly e) It depends on pressure only

The solubility of a gas in a solvent/solution (including pure water as a solvent) will **decrease**. There are two reasons:
i) the dissolving of a gas molecule in a liquid (among water molecules) causes a release of heat energy (it is exothermic). Later in the course, we will learn something called Le Chatelier's Principle which explains how perturbations to an equilibrium occur: in an exothermic reaction, energy is released, and the temperature would fall in the system, but by adding heat energy to a system (increasing the temperature), it shifts the reaction of gas dissolution in the opposite direction, and so the gas does not want to go from the gas to liquid ("dissolved") phase.
ii) here is a more intuitive way of seeing it: by heating up the water or any solution (implied by temperature increase) all particles are given more kinetic energy (the velocity of the molecule increases). With (gas) molecules moving faster, they can break the intermolecular forces that hold them in the solvent and escape to gas phase.

25. How many moles are in 25.0 g of NaCl? (Use the Periodic Table to determine NaCl molar mass)

a) 2.34 mol
b) 0.427 mol
c) 1.00 mol
d) 0.234 mol
e) 0.584 mol

This question really does not belong in this midterm. It is a Lecture Set 5 question, where the concept of calculating moles from masses was addressed. There was a serious concern about the students learning the basic concepts, so this question was thrown in. At any rate, the calculation is ***mass (g) / molar mass (g/mol)*** to give the result in moles: (25.0 g)/(58.44 g/mol) = 0.427789 mol → 0.428 mol. This is not one of the choices that is the correct answer given the precision, but (b) is nonetheless the correct answer because the exam asks for the best choice, which this is.

Continue to next page for True-False questions

26. A decomposition reaction involves two compounds combining to form a larger complex product
   a) true b) **false**

False. This is opposite of decomposition and looks like a combination.

27. The oxidation number of a pure element is always zero.
   a) **true** b) false

This actually one of the oxidation number rules (the first one)

28. A strong electrolyte completely dissociates into ions in aqueous solution.
   a) **true** b) false

Basically the definition of a strong electrolyte

29. The molarity of a solution is expressed in units of **mol/kg**.
   a) true b) **false**

These are the units of molality, not molarity. Units of molarity are **mol/L** or ***M***.

30. A precipitate is an insoluble compound (a substance forming a solid from solution).
   a) **true** b) false

The definition of precipitate actually.

31. The equation $PV = nRT$ is known as the Ideal Gas Law equation.
   a) **true** b) false

That is the equation, one and the same.

32. In chemical reaction equation, compounds to ***left*** of the ***reaction arrow*** are called **products**.
   a) true b) **false**

Reactants to left of arrow, products to the right of the arrow.

33. Ionic compounds prefer to dissolve in polar solvents like water, not in nonpolar solvents.
   a) **true** b) false

The like dissovles like concept

34. Electronegativity is unrelated to an element's tendency to gain or lose electrons.
   a) true b) **false**

Electronegativity is the pull of electrons forming a bond that an element has in that bond with another element, so it is related to oxidation number actually.

35. At Standard Temperature and Pressure (STP), the temperature is 0°C and the pressure is 1 atm.
   a) **true** b) false

The very definition of STP.